

Cloud Kitchen Market Analysis & Data Strategy Report

1. Project Objective

The primary goal was to analyze restaurant data from food delivery platforms (Foodmandu/Bhoj) to identify market "gaps" in Kathmandu. We aimed to determine the best location and pricing strategy for a new Cloud Kitchen to maximize profit and minimize competition.

2. The Data Pipeline Process

Step 1: Data Cleaning & Preparation

Raw data is often "noisy." We performed the following cleaning steps to ensure mathematical accuracy:

- Price Formatting:** Converted currency strings (e.g., "1,190.00") into floating-point numbers.
- Location Normalization:** Simplified specific addresses into broader neighborhood areas (e.g., "Putalisadak").
- Anomaly Detection:** Identified and removed extreme outliers. While some items were listed above 100,000 NPR (likely typos or bulk packages), we applied a domain-knowledge filter to focus on standard meal prices under 20,000 NPR.

Step 2: Feature Engineering

To turn raw data into business intelligence, we created three new metrics:

Feature	Business Purpose
Price Tier	Categorized meals into 'Budget', 'Mid-Range', and 'Premium' to understand market segments.
Menu Size	Calculated the total items per restaurant to measure operational complexity.
Area Competition Count	Counted unique restaurants per neighborhood to identify market density.

Step 3: Exploratory Data Analysis (EDA)

Using **Seaborn** and **Matplotlib**, we uncovered critical market trends:

- The Price Curve:** The market is "right-skewed," with 75% of food items costing less than 675 NPR.
- The Mid-Range Trap:** Data visualization proved that the "Mid-Range" tier is the most crowded, with the highest average density of competitors.

Step 4: Strategic Gap Analysis (The 4-Quadrant Map)

We built a proprietary 2D map comparing *Competition Density* vs. *Average Price*. This allowed us to categorize neighborhoods into four types:

1. **The Gold Mine:** High Price tolerance, Low Competition (Target Area).
2. **The Bloodbath:** Low Price, High Competition (Avoid).
3. **Premium Saturated:** High Price, High Competition.
4. **Low-Value Desert:** Low Price, Low Competition.

Step 5: Preprocessing for Machine Learning (ML-Ready)

Finally, we prepared the dataset for advanced predictive modeling by a Data Science team:

- **Ordinal Encoding:** Converted 'Price Tier' into numerical values (0, 1, 2) to preserve the price hierarchy.
 - **Feature Scaling:** Used StandardScaler to normalize 'Price', 'Menu Size', and 'Competition Count' to a standard scale, preventing any single variable from dominating the model's math.
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3. Final Recommendation

The analysis suggests that a new Cloud Kitchen should avoid the highly saturated Mid-Range market and instead focus on the **Gold Mine** neighborhoods identified in our quadrant map, utilizing a lean menu strategy to compete with large, slower-moving legacy restaurants.